

A lightweight multi-modality medical image semantic segmentation network base on the novel UNeXt and Wave-MLP

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ABSTRACT

Medical images sometimes contain diseased regions that are different sizes and shapes, which makes it difficult to accurately segment these areas or their edges. However, directly coupling CNN and MLP to construct global and local dependency models may also cause significant computational complexity issues. In this paper, a unique, lightweight UNeXt network segmentation model for medical images based on dynamic aggregation tokens was proposed. Firstly, the Wave Block module in Wave-MLP was introduced to replace the Tok-MLP module in UNeXt. The phase term in Wave Block can dynamically aggregate tokens, improving the segmentation accuracy of the model. Secondly, an AG attention gate module is added at the skip connection to suppress irrelevant feature representations in the sampling path of the encoding network, thereby reducing computational costs and paying attention to noise and artifacts. Finally, the Focal Tversky Loss was added to handle both binary and multiple classification jobs. Quantitative and qualitative experiments were conducted on two public datasets: COVID-19 CT and BraTS 2018 MRI. The Dice score, Precision score, recall score, and Iou score of the proposed model on the COVID-19 dataset were 0.928, 0.867, 0.916, and 0.940, respectively. On BraTS 2018, the Dice scores of the ET, WT, and TC categories were 0.933, 0.925, and 0.918, respectively, and the HD scores were 1.595, 2.348, and 1.549, respectively. At the same time, the model is lightweight and has a considerably decreased training time with GFLOPs and Params of 0.52 and 0.76, respectively. The proposed lightweight model is superior to other existing methods in terms of segmentation accuracy and computing complexity according to experimental data.

1. Introduction

To diagnose and treat cancers, medical scanning images are essential for identifying tumors and calculating tumor sizes in various organs. For example, the analysis of lung CT scanning images is very convenient for detecting lung cancer and COVID-19 disease. It will well reflect can be further paired with other imaging methods (such as ultrasound) to provide doctors with organ visualization during surgeries. However, radiologists with years of experience must spend a lot of time manually segmenting medical images.

The following are the practical research implications of this article:

- (1). The accurate segmentation of medical images can provide doctors with detailed disease information, such as tumor and infection area size, to provide a basis for further treatment.
- (2). Using artificial intelligence for large-scale batch diagnosis of CT images can save radiologists time and reduce misdiagnosis rates.

Medical images can be automatically segmented swiftly and precisely with the help of artificial intelligence systems, thereby replacing manual segmentation and facilitating auxiliary diagnosis. These automated segmentation algorithms can help surgeons save time and effort so that they can concentrate more on the actual operation and therapy, but automatic and accurate segmentation of medical images is very challenging (Muhammad et al., 2022). Due to their superior efficiency and sophisticated design, UNet (Ronneberger et al., 2015) and its variants (Zhou et al., 2018a; Huang et al., 2020; Lou et al., 2021) are frequently used in medical image segmentation tasks. Unet (Ronneberger et al., 2015) is an encoder-decoder architecture that uses skip connections to aggregate features from the encoder with the up-sampled features of the decoder to produce high-resolution segmented images. Huang et al. (2020). recommend using full size skip connections, including internal connections between decoder blocks. DC-UNet, which makes the use of residual pathways in skip connections and

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multi-resolution convolutional blocks, was developed by Lou et al (He et al., 2016). As computer vision has advanced, the ResNet architecture has been widely adopted as the foundation for medical image segmentation. Also used for pathological alterations and organ segmentation (Feng et al., 2020; Gu et al., 2019) are pyramid pools and dilated convolutions (Chen et al., 2017). Inspired by ViT, the introduction of Transformer to medical image segmentation tasks also shows great advantages (Chen et al., 2021; Cao et al., 2021; Dong et al., 2021). Chen et al. (2021) proposed TransUnet, which combines CNN and Transformer to build an encoder to capture global features. For the extraction of local features, CNN is used as a decoder to extract detailed local features. Li et al. (2022) proposed that TFCNs can filter out non-semantic features. Cao et al. (2021) proposed Swin-Unet.

In a variety of computer vision applications, particularly in the segmentation of medical images, UNet (Ronneberger et al., 2015) and its variants (Zhou et al., 2018b; C. i et al., 2016; Oktay et al., 2018a) and Transformer-based approaches have produced excellent results. However, this type of network may generate a huge amount of computation, making it difficult to easily apply to clinical diagnosis. To improve the performance of models, many scholars generally stack multiple Transformer blocks or add a self-attention mechanism to the Transformer network, resulting in $O(n^2)$ temporal and spatial complexity. To compress models, many lightweight models have been proposed for better application in practice to facilitate diagnostic assistance.

(1) Lightweight models based on Vision Transformers (ViT).

The basic ViT model has a huge number of parameters, and some studies have further compressed the model through some strategies. Touvron et al (Chen et al., 2021) proposed DeiT, attempting to change the training strategy of the model to save training costs. Wang et al. (2021) proposed a Pyramid Vision Transformers (PVT) that uses a lightweight attention mechanism in space, thereby saving training time. In PVTv2, Wang et al. (2022) introduced LSRA, which has linear complexity and improves the performance of the model. Furthermore, Chen et al. attempted to combine CNN and Transformers to create a series of models such as MedT (Valanarasu et al., 2021) and UNETR (Ali et al., 2022a).

(2) Lightweight models based on attention mechanisms.

The context-axial attention network, according to Chen et al (Ange et al., 2022), decreases the weight of the model while also enhancing its accuracy. Wang et al (Sanghyun et al., 2018) proposed a Convolutional Block Attention Module (CBAM) that focuses on location information in the feature map from both channel and spatial dimensions, and the computational complexity of the CBAM module is very small. SegFormer (Yutong et al., 2021) introduces a deformable self-attention mechanism to fully utilize feature extraction for local details. UNet (Yunhe et al., 2021) innovatively applies self-attention modules to encoders and decoders, thereby achieving relatively excellent performance with fewer parameters. DS-tranunet (Ailiang et al., 2021) proposes a TIF module that utilizes the MSA mechanism to achieve high performance while compressing the model. TransAttUnet (Bingzhi et al., 2021) created a multiscale skip connection as well as a multilevel guided attention network. MissFormer (Xiaohong et al., 2021) redesigned the feedforward network to allow contextual modeling of global and local features.

Mt-UNet (Hongyi et al., 2022) designed a locally global Gaussian weighted self-attention (LGG-SA) to effectively calculate self-correlation. The goal of SAM is to create a universal tool for subdividing various medical images. To adapt SAM to medical image segmentation, MedSAM (Junde et al., 2023) is an improvement based on the SAM model.

To address the limitations of CNN model in global representation, the ViT (Alexey et al., 2020) model is proposed, which uses multiple self-attention (MSA) mechanisms to model remote dependencies. This method first decomposes the image into token sequences and then embeds the position into the token sequence when the token sequence is fed into the Transformer block. Compared to other convolution-based methods, it achieves SOTA performance.

Table 1 summarizes the medical image segmentation networks in recent years. After summarizing and analyzing the literature, it is found that there are still two main shortcomings in medical image segmentation.

1.1. Shortcoming 1

Due to the different size and shape of the lesion areas in the image, erroneous segmentation or inaccurate edge segmentation often occurs.

Deep and shallow features in medical images are crucial for image segmentation. Deep features are conducive to capturing global semantic information and accepting large receptive fields, while shallow features are conducive to extracting small features and accepting small receptive fields. Using CNN alone, due to its limited receptive field, may pay more attention to extracting of local features, which is not conducive to global information extraction, and may result in incorrect segmentation, as shown in Fig. 1. In the upper left figure of Fig. 1, using CNN methods for segmentation may cause incorrect segmentation problems, such as when there is one lesion area in Ground truth while using the Unet++ network to segment lung lesions in three blocks. In the lower left image of Fig. 1, using the Unet++ did not segment the dark gray dot-like tumors in the white areas of brain tumors. While the Transformer-based method can well preserve the global semantic information of segmentation features, local detail segmentation is not accurate, as shown in Fig. 1 Challenge 2. In the upper right of Fig. 1, although there are no missed or incorrect segments, the segmentation results for some details are still not ideal. For example, in brain tumor segmentation, the small dark gray area inside the white color is not segmented using TransUnet. In the lower right of Fig. 1, there is a difference between the TransUnet segmentation result and the Ground truth segmentation result for the small area edge of COVID-19 disease pneumonia circled in red.

1.2. Shortcoming 2

Directly coupling CNN and Transformer to build global and local dependency models can lead to significant computational complexity.

Due to the heavy parameters and complex calculations of some networks and the fact that most medical images are three-dimensional data, some existing networks cannot be effectively used in practice. If it can improve the efficiency of the model and better apply it to practice,

Table 1
Some mainstream medical image segmentation networks.

Author	Model	Innovation direction	Params (M)	Evaluation indicators
Xie et al. (2021)	SegFormer	Transformer with MLP	B4:64.1	ADE20K Iou is 0.511
Wang et al. (2022)	Pyramid Vision Transformer	Transformer	32.1	mIou is 0.426
Ali et al. (2022a)	UNETR	Transformer with CNN	92.58	BTCV Dice is 0.899
Yan et al. (2022)	CaraNet	CNN	–	Synapse Dice is 0.786
Wang et al. (2022)	Mt-UNet	Transformer with CNN	–	VC-T mean Dice is 0.903
Wu et al. (2023)	MedSAM	Transformer	–	Brain-Tumor Dice is 0.876

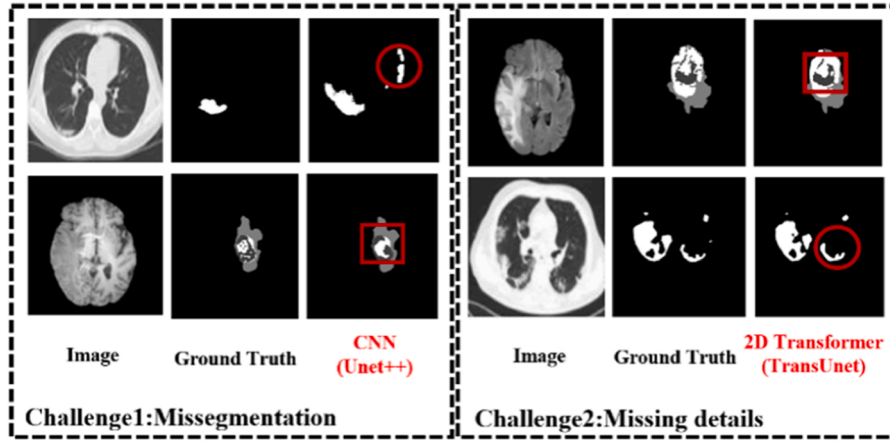


Fig. 1. The main existing challenges of medical image segmentation. Challenge 1: Some CNN-based methods may encounter segmentation errors. Challenge 2: Transformer-based methods can well preserve the global semantic information of segmentation features, but local detail segmentation is not accurate.

it will reflect its practical application value well. Therefore, both model efficiency and segmentation accuracy are equally important, and it is necessary to develop lightweight image segmentation models that have enough receptive fields while paying attention to local feature details.

Many of the networks in Table 1 combine Transformer with CNN or MLP (Tolstikhin et al., 2021; Hao et al., 2022; Xiaohan et al., 2021; Kyriazi and Luke, 2021), but the model parameters are relatively large. Recently, some MLP-based networks (Kyriazi and Luke, 2021; Menze et al., 2015; Bakas et al., 2017; Yehui et al., 2022; Rahman and Radu, 2023; Valanarasu et al., 2022; Jonathan et al., 2019; Ze et al., 2021) have been proposed for medical image segmentation, with performance comparable to some CNN networks and relatively small model parameters. MLP Mixer (Kyriazi and Luke, 2021) is a pure MLP structure that achieves feature fusion in both token and Channel dimensions. Therefore, this article attempts to explore the combination of CNN and MLP to propose a lightweight medical image segmentation network based on CNN and MLP. The proposed method was improved based on the UNeXt network, which is the first lightweight network based on MLP. The improved network follows the CNN structure of the UNeXt encoder and decoder, replacing Tok-MLP with Wave-MLP in the original model. Due to the introduction of fewer inductive biases in Wave-MLP, the generalization ability of the model is relatively strong, and this concise structure will bring less computational complexity. Experiments on the COVID-19 CT and BraTS 2018 MRI datasets (Menze et al., 2015; Bakas et al., 2017) have verified the excellent performance of the proposed model. More importantly, compared to Transunet, it reduces the number of parameters by 99.28% and the computational complexity by 98.11%, making it suitable for medical imaging applications. The main contributions to this paper are as follows:

(1). The introduction of the Wave Block module in Wave MLP (Yehui et al., 2022) replaced the tokenized MLP (Tok MLP) module in UNeXt. The phase terms in Wave Block can dynamically aggregate tokens, which not only improves the segmentation accuracy of the model but also compresses it, achieving a balance between model accuracy and computational complexity.

(2). The AG (Rahman and Radu, 2023) module attention gate was introduced at the skip connection to suppress irrelevant feature representations in the sampling path of the encoding network, allowing the model to retain more relevant spatial information, thereby reducing computational costs and reducing attention to noise and artifacts.

(3). A new Loss function Focal Tversky Loss (FTL) was added. FTL attempts to use coefficients to learn difficult examples, such as in situations where the ROI (region of interest) is small, to solve the problem of class imbalance in medical image datasets.

The structure of this paper is as follows: Section 2 mainly introduces the UNeXt (Valanarasu et al., 2022) model and the innovative work of

this paper. Section 3 describes the basic setup, datasets, and qualitative and quantitative results of the experiment. Section 4 discusses ablation experiments, model complexity experiments, training strategy experiments, confidence experiments, and training process experiments. Section 5 summarizes the research findings and future work.

2. Methods

As shown in Fig. 2, a lightweight multi-modality medical image semantic segmentation network was conducted. Firstly, the data is augmented through translation, rotation, contrast enhancement, and other expansion methods. Then, the augmentation data is input into the improved UNeXt model, and after training and testing, a segmentation image is obtained. Finally, morphological open operations are used to refine the edges of the segmented image.

2.1. UNeXt network

Like the Unet model, UNeXt is also a U-shaped structure that includes a convolution stage and an MLP stage. The input image passes through the encoder. The first three modules are convolutional modules, and the

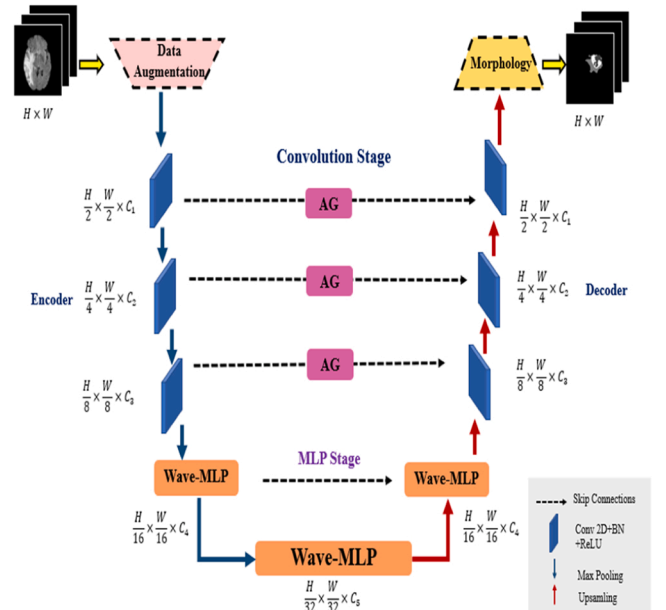


Fig. 2. Overall framework diagram of the improved UNeXt network.

next two are Tok-MLP. The decoder has two Tok-MLP modules, followed by three CNN modules. The image first extracts local features through CNN down-sampling, then performs global modeling of the features through the MLP phase. Finally, the image enhances the extraction of local features through CNN up-sampling, with intermediate skip connections to compensate for feature loss during the down-sampling process.

Convolution phase: Convolution blocks are composed of convolutional layers, BN, and ReLU activation functions. The kernel of the convolution is 3×3 . Set the stride to 1 and the padding to 1. Up-sampling uses traditional Max Pooling after convolution, which can preserve features while reducing the amount of computation and preventing model overfitting to a certain extent. Down-sampling uses bilinear interpolation instead of Transposed Convolution, because Transposed Convolution has a checkerboard effect and the segmentation boundary is jagged.

Fig. 3(a) shows the Tok-MLP structure of the original UNeXt network. The features extracted from the convolution first pass through the shifted MLP (width direction) and then through the GLUE+Encoding combination. GLUE is used instead of ReLU because GLUE can achieve better performance in smoothing. After that is the shifted MLP (height direction). Because Tok-MLP has two MLPs, the extracted physical signs can move in width and height, like the axial attention mechanism (Jonathan et al., 2019). The shifted MLP structure in Tok-MLP is shown in Fig. 3(b), introducing a window-based attention mechanism to build a network model suitable for global and local feature extraction and fusion. This method is like Swing Transformer (Ze et al., 2021).

2.2. Novel UNeXt and Wave-MLP network

2.2.1. Introduction of Wave-MLP module

The improved model introduces the Wave Block module in Wave-MLP, replacing the Tok-MLP module. The phase term in Wave Block can dynamically aggregate tokens to improve the segmentation accuracy of the model while compressing it.

The parameters in the UNeXt network mostly come from the MLP stage, so to further lighten the model and reduce model parameters, we attempt to improve the MLP stage. Different Vision MLPs have different aggregation methods for tokens, mixing the fixed weights of the full connection layer with different tokens. Transformer (Rahman and Radu, 2023) is designed to aggregate tokens and compute the inner product between various tokens using dynamically changed weights. The greater the similarity, the more weight tokens have in the aggregation process. Other MLPs also heap up token FC layers and activation functions to

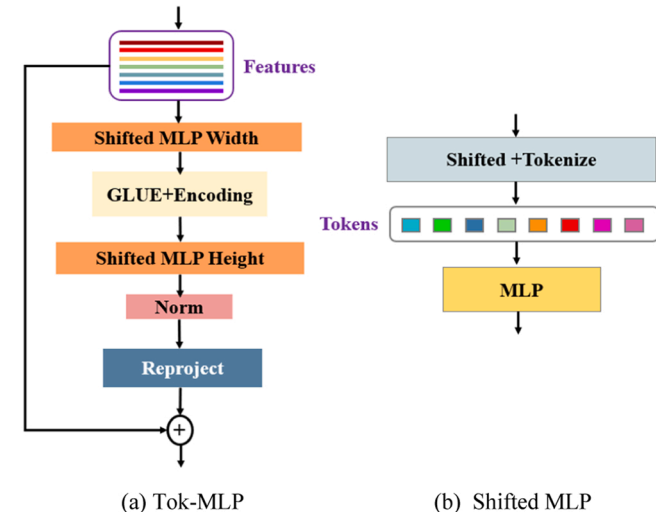


Fig. 3. Framework (a) Tok-MLP (b) Shifted MLP.

build hybrid MLPs, such as LPMixer (Tolstikhin et al., 2021) and ResMLP (Hugo et al., 2022). Unlike Transformer, which improves the attention mechanism, this paper aims to improve the representation of tokens by dynamically aggregating tokens based on their semantic content.

In quantum mechanics, entities (such as electrons and photons) are typically represented by wave functions that include amplitude and phase (such as De Broglie waves). Inspired by quantum mechanics, token is used to describe waves with amplitude and phase that can dynamically polymerize. Phase is the complex value of the connection between the modulated token and the fixed weight in the MLP, whereas amplitude is the distinctive real value of each token's content. Wave-MLP adds a Phase Aware token Mixing (PATM) module to dynamically estimate the phase of each token since the semantic information in tokens varies depending on the input image.

Given the input characteristics x_j , amplitude z_j , and phase θ_j are the channel FC and phase estimation modules, respectively. In Wave-MLP, token is represented as a Wave- $\tilde{z}_j$ with amplitude and phase information, namely:

$$\tilde{z}_j = |z_j| \odot e^{i\theta_j}, j = 1, 2, \dots, n \quad (1)$$

$\odot$ is the element direction multiplication amplitude, and z_j is the characteristic real value of the aggregated token. θ_j represents the phase.

The Wave-MLP framework diagram as shown in Fig. 4 mainly includes two key modules: (1) Token Mixing, which considers amplitude and phase information to aggregate different tokens. (2) Channel MLP is used to extract the characteristics of each token. token Mixing consists of Channel-FC, nonlinear activation function, and PATM, and the nonlinear activation function is used to extract the characteristics of each token. Eq. 2 describes Channel-FC in token Mixing.

$$\text{Channel-FC}(z_j, W^c) = W^c z_j, j = 1, 2, \dots, n \quad (2)$$

Where W^c is the parameter. Channel-FC independently operates on each token to facilitate feature extraction. Represent an intermediate feature $Z = [z_1, z_2, \dots, z_n]$ containing n markers, where each marker z_j is a dimensional vector.

Two concurrent PATM modules in the model combine spatial data in the high and broad dimensions, respectively. Several branches are added together using a reweighting module, like CycleMLP.

2.2.2. Introducing the AG Attention Gate

Introducing AG attention gates at UNeXt skip connections suppresses irrelevant feature representations in the sampling path of the encoding network, allowing the model to retain more relevant spatial information, thereby reducing computational costs and reducing attention to noise and artifacts. The AG module is shown in Fig. 5 below, and its principle is to suppress features in unrelated background regions through channel signals based on the image spatial information (Oktay

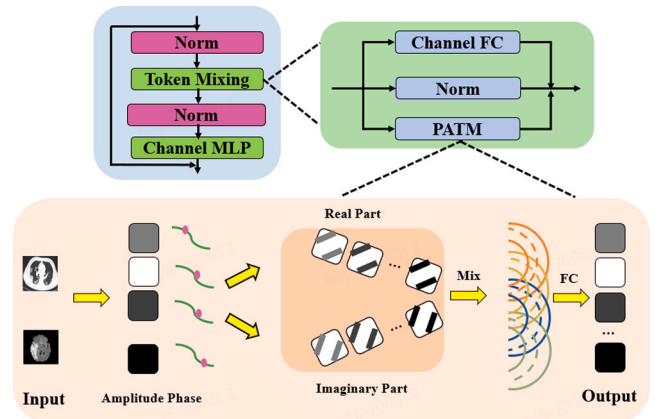


Fig. 4. Novel Wave-MLP architecture diagram.

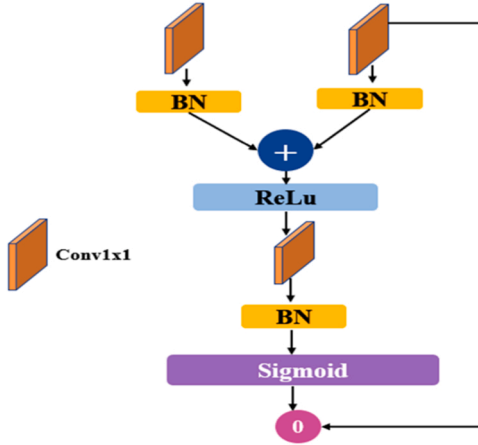


Fig. 5. AG module.

et al., 2018b). Specifically, channel signals are used to aggregate multi-level features of skip connections, which increases the ability of feature extraction to extract spatial information. Like Attention UNet, additive attention is used to obtain gating coefficients, as relevant studies have shown that using additive attention can achieve efficient context modeling with linear complexity. Eqs. 5 and 6 provide the additive attention gate AG:

$$q_{att}(g, x) = \sigma_1(BN(C_g(g) + BN(C_x(x)))) \quad (3)$$

$$AG(g, x) = x * \sigma_2(BN(C(q_{att}(g, x)))) \quad (4)$$

Where $\sigma_1(\cdot)$ and $\sigma_2(\cdot)$ correspond to ReLU and S-shaped activation functions, respectively. $BN(\cdot)$ is a batch normalization operation. C and x are up-sampled and skip connection features, respectively.

2.2.3. Introducing a new Loss function Focal Tversky Loss

Due to the frequent occurrence of category imbalance in annotated images, they may be interfered with pixels (background) with more pixels during training, while smaller feature pixels may not be able to learn. Therefore, this paper adds a new loss function, Focal Tversky Loss (FTL), which attempts to use the coefficients to learn difficult examples, such as in situations where the ROI (Region of Interest) is small, to solve the problem of category imbalance in medical image datasets.

(1) Dice loss is one of the most commonly used segmentation indicators. Unlike cross-entropy, it does not require the reweighting of unbalanced segmentation tasks.

$$L_{Dice} = 1 - \frac{2|X \cap Y|}{|X| + |Y|} \quad (5)$$

$|X \cap Y|$ represents the number of elements with the intersection of X and Y sets. For segmentation tasks, $|X|$ and $|Y|$ represent the number of elements for the ground truth and predict the mask, respectively. (2) Minimizing HD directly is very difficult. Karimi et al. proposed an HD method for Ground truth estimation and prediction. Using the following HD loss function can reduce HD and be used for training.

$$L_{HD} = \frac{1}{N} \sum_{i=1}^N [(S_i) - g_i \times (d_{G_i}^2 + d_{S_i}^2)] \quad (6)$$

$d_{G_i}^2$ and $d_{S_i}^2$ is the distance transformation of ground truth and segmentation.

(3) Focal Tversky Loss is like Focal Loss and attempts to use the coefficients to learn difficult examples, such as when the ROI is small.

$$L_{TL} = \frac{\sum_{i=1}^N p_{ic} g_{ic} + \epsilon}{\sum_{i=1}^N p_{ic} g_{ic} + \alpha \sum_{i=1}^N p_{i\bar{c}} g_{ic} + \beta \sum_{i=1}^N p_{ic} g_{i\bar{c}} + \epsilon} \quad (7)$$

Among them p_{ic} is the probability that pixel i belongs to category c of the lesion, $p_{i\bar{c}}$ is the probability that pixel i belongs to the non-pathological category, g_{ic} and $g_{i\bar{c}}$ similar. In cases of imbalance, hyper-parameters can be adjusted by α and β .

$$L_{FTL} = \sum_c (1 - Tl_c)^{\gamma/1} \quad (8)$$

γ Varies within the range (Muhammad et al., 2022; Zhou et al., 2018a).

The network is designed to segment COVID-19 CT images and multi-segment the BraTS dataset. Because the UNet network is aimed at the skin for segmentation, if the original model loss function is used to segment the BraTS dataset, the result is not ideal, so this paper added a new loss function. The following is a new loss function after the change.

$$L_{DF} = 0.5 \log L_{Dice}^2 + \log L_{HD}^2 + \log L_{FTL}^2 \quad (9)$$

3. Experimental results

In this section, we first describe the experimental settings, introduce two public datasets, namely, 2D COVID-19 CT images segmentation and 3D BraTS 2018 MRI images segmentation, and introduce the evaluation indicators for medical image segmentation. Next, the performance of the proposed method was analyzed through a series of quantitative, qualitative, and ablation experiments.

3.1. Experimental settings

Using the NVIDIA V1032G memory training model and the PyTorch neural network library for development, Evaluate the model on a local workstation. The images were set to 224×224 pixels. For segmentation of COVID-19 lesions, learning rate η is set to 0.05; for BraTS brain tumor segmentation, η is set to 0.01. Using the ADAM optimizer training model for both datasets, batch_size is set to 16 and the number of epochs is 500. ReLU is selected as the Activation function.

3.2. Dataset

(1) COVID-19 CT Scan Image Dataset.

The public dataset of was used, including 20 annotations of the chest CT volume of COVID-19. A three-dimensional dataset has 160 images, each with a size of 224×224 . This paper uses CT scan images of infected COVID-19 for training testing.

(2) BraTS 2018 MRI Brain Tumor Image Dataset.

The dataset was constructed by the Biomedical Imaging Research Group (CBIA) of the University of Pennsylvania. It contains multimodal brain MRI scans and is manually annotated with real tags from 285 patients with glioblastomas. The 285 cases in BraTS are further divided into HGG (210 cases) and LGG (75 cases). HGG is a high-grade glioma (WHO 3–4 grades), and LGG is a low-level glioma (WHO 1–2 grades). An MRI sequence has 155 images, each with a size of 224×224 . There are four brain MRI modes, as shown in Fig. 6. Flair, T1, T1ce, and T2 can be used for prediction. As shown in Fig. 7, there are three brain tumor regions: ET (Enhanced Tumor Blue), WT (Whole Tumor Green), and TC (Tumor Core Red). In this study, T1ce was selected as the input image for this article, and the real label uses the NET type because it depicts the smallest area for small object segmentation.

3.3. Evaluation metrics

The evaluation indicators in this paper are Dice similarity coefficient, Precision, Recall, Spec, and Iou. The expression is defined as follows:

$$\text{Dice} = \frac{2 * \text{area}(S_1 \cap S_2)}{\text{area}(S_1) + \text{area}(S_2)} \quad (10)$$

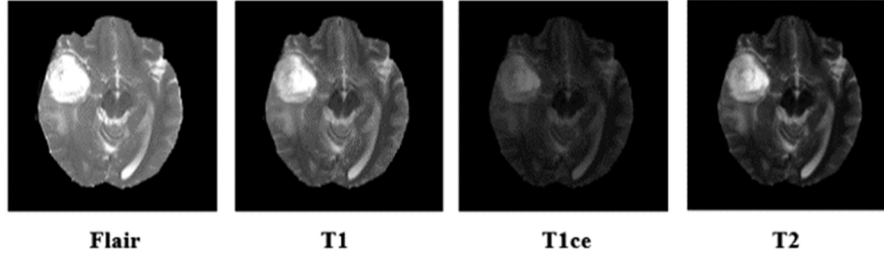


Fig. 6. Examples of Flair, T1, T1ce, and T2.



Fig. 7. ET (Enhanced tumor, blue), WT (Whole tumor, green), and TC (Tumor core, red).

$$Precision = \frac{TP}{TP + FP} \quad (11)$$

$$Recall = \frac{TP}{TP + FN} \quad (12)$$

$$Spec = \frac{TN}{TN + FP} \quad (13)$$

$$Iou = \frac{area(S_1 \cap S_2)}{area(S_1 \cup S_2)} \quad (14)$$

S_1 and S_2 are the pixels of infected COVID-19 lesions (brain tumor pixels) predicted by the model and annotated by experts, respectively. In this model, positive and negative refer to pixels of COVID-19 lesions (brain tumors) and non-COVID-19 lesions (non-brain tumors), respectively. TP indicates that pixels of COVID-19 lesions (brain tumors) are correctly detected; TN indicates that pixels of non-COVID-19 lesions (non-brain tumors) are correctly detected as pixels of non-COVID-19 lesions (non-brain tumors); FP indicates that pixels of non-COVID-19 lesions (non-brain tumors) are incorrectly detected as pixels of COVID-19 lesions (brain tumors); FN indicates that pixels of COVID-19 lesions (brain tumors) are incorrectly detected as pixels of non-COVID-19 lesions (non-brain tumors).

In this paper, the Hausdorff Distance (HD) is used to calculate the similarity between the test results and the images annotated by experts. The definition is as follows:

$$HD = \max\{\max_{a \in A} \min_{b \in B} \|a - b\|, \max_{b \in B} \min_{a \in A} \|b - a\|\} \quad (15)$$

a is the predicted infected COVID-19 lesion (brain tumor) pixel, and b is the expertly annotated COVID-19 lesion (brain tumor) pixel.

3.4. Data augmentation

Data augmentation includes adjusting the image size using a ratio of 0.6, 1.2, 1.5, rotating by 30, 45, 90, 120, 180, 270 degrees, translating by (15, 15), (−15, 15), (15, −15), and (−15, −15) pixels, enhancing the image brightness by using an enhancement factor of 0.5, 0.7, and enhancing the image chroma by using an enhancement factor of 0.9, 1.3 and 1.5. Twenty 3D volumetric images were sliced from the COVID-19 infection. After each slice, the lesion's pixel size was divided into five categories. One was selected from each of these five types, for a total of

100 2D images. To prevent model overfitting, image augmentation methods were used to augment 100 CT images to 2000. For the BraTS dataset, 200 3D volumetric HGG images were chosen for slicing and classifying into two groups based on ET pixel size. After each slicing, two 2D images were selected, a total of 400 2D images with annotations were selected, and the 400 CT images were augmented to 4000 using image augmentation methods. Both datasets use 80% for training and 20% for testing.

3.5. Experimental results

3.5.1. Experimental results on the COVID-19

Table 2 describes the quantitative comparison results of different methods on the COVID-19 dataset. The selected comparison models are divided into four categories.

- 1) CNN-based method: using the Unet network proposed by Ronneberger in 2015; the Unet++ (Zhou et al., 2018b) network has made some improvements based on the Unet network. This type of network segmentation has strong versatility, but for some relatively difficult lesions or tumors, the results obtained by segmentation are not great. For example, the examples cited in the citation may lead to problem of missegmentation and missing detail, and the number of model parameters is relatively large.
- 2) 3D Transformer, because most medical images are three-dimensional. 3D images segmentation can directly segment 3D images, with a certain degree of performance improvement. However, the model parameters are also huge, and they are still impossible to achieve due to insufficient computational power.
- 3) 2D Transformer, which has become very popular for image segmentation in recent years. Some mainstream networks, such as Swin-Unet, TransUnet, HiFormer, MedSAM and others, have shown good performance on some medical image segmentation datasets.

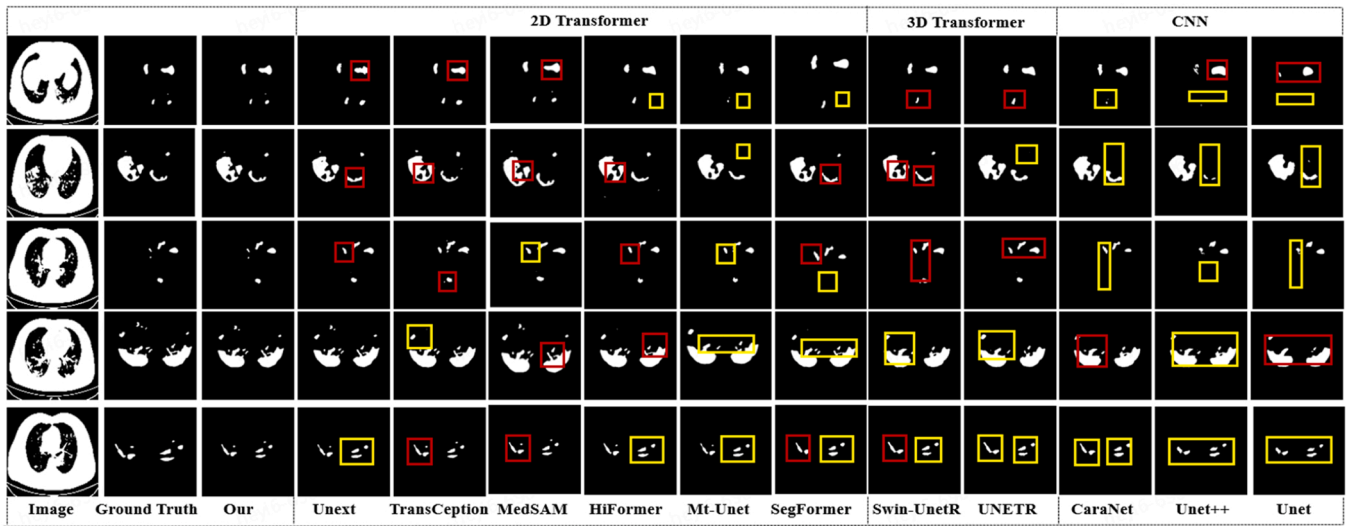
This paper is an improvement on the UNeXt network. After the improvement, the network's performance has greatly improved. Compared to the previous best UNeXt, the proposed method has an average increase of 3.9%, 6.4%, 3.5%, and 4.3% in Dice, Iou, Recall, and Precision. Compared to some other 2D Transformer networks, the parameters and GFLOPs have been significantly reduced, and the UNeXt network has reached a parameter amount of 1.47. Compared to it, the network parameters of the proposed method have been reduced by 8.7%, and the computational complexity has been reduced by 48.2%, with the least number of parameters, indicating the performance and efficiency of the proposed network.

Fig. 8 depicts a sample comparison of five experimental groups. As shown in Fig. 8, the proposed method can effectively solve the problem of missed segmentation in some CNN-type networks. For example, in the first line, Ground truth has four small lesion areas, while Unet misses two small lesion areas in the yellow box, and Unet++ misses one small lesion area in the yellow box. Because the network in this paper is a CNN combined with an MLP network, which not only focuses on local details but also on global context modeling, it can effectively avoid erroneous

Table 2

Quantitative comparison of performance and parameter indicators for the model on the COVID-19 dataset.

Model Type	Method	Year	Dice↑	Iou↑	Recall↑	Precision↑	P	GFLOPs↓	Params (M)↓
CNN	Unet (Ronneberger et al., 2015)	2015	0.711	0.552	0.656	0.778	0.031	57.84	33.24
	Unet++ (Zhou et al., 2018b)	2015	0.727	0.571	0.680	0.781	0.020	37.23	40.13
	CaraNet (Ange et al., 2022)	2022	0.785	0.646	0.743	0.832	0.005	49.53	57.42
3D Transformer	UNETR (Ali et al., 2022a)	2022	0.735	0.581	0.692	0.784	0.030	84.76	92.79
	Swin-UNetR (Ali et al., 2022b)	2022	0.754	0.605	0.764	0.744	0.042	77.34	62.18
	SegFormer (Yutong et al., 2021)	2021	0.718	0.560	0.671	0.772	0.042	95.70	64.10
	TransUnet (Muhammad et al., 2022)	2021	0.773	0.630	0.768	0.779	0.019	27.54	105.82
	Swin-Unet (Cao et al., 2021)	2021	0.779	0.638	0.785	0.775	0.009	15.67	21.43
2D Transformer	PVT (Wang et al., 2021)	2022	0.693	0.530	0.612	0.799	0.021	31.60	32.10
	HiFormer (Moein et al., 2023)	2022	0.744	0.592	0.746	0.742	0.008	19.54	25.51
	Mt-Unet (Hongyi et al., 2022)	2022	0.789	0.652	0.751	0.831	0.010	24.31	64.26
	MedSAM (Junde et al., 2023)	2023	0.811	0.682	0.761	0.868	0.006	105.32	307.93
	TransCeption (Reza et al., 2023)	2023	0.814	0.686	0.792	0.837	0.006	27.43	38.14
	UNeXt (Valanarasu et al., 2022)	2022	0.889	0.803	0.881	0.897	0.005	0.57	1.47
CNN+MLP	Our	–	0.928	0.867	0.916	0.940	–	0.52	0.76

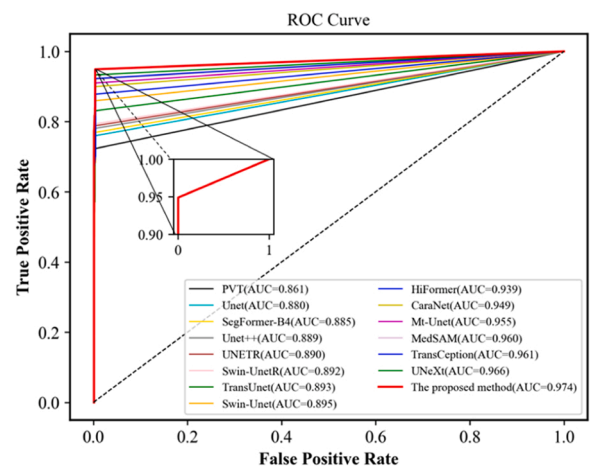
**Fig. 8.** Comparison of model segmentation results on the COVID-19 dataset (The red box indicates a detail segmentation error, and the yellow box indicates a missing segmentation).

segmentation and missed segmentation. Compared to the 3D Transformer method, 3D Transformer may not achieve ideal segmentation results for some small lesions, such as the last line, Swin-UnetR (Ali et al., 2022b), and UNETR, which cannot segment some adhesive lesions well.

For some 2D Transformer networks where edge segmentation is not ideal, this model adds an AG module at the skip connection to reduce information loss during feature fusion. Edge segmentation can also achieve good results. As shown in the fourth line, TransCeption, HiFormer and MedSAM have the problem of missing segmentation errors for small independent lesions. While the UNeXt network may have erroneous segmentation for small lesions, such as the third line, where the segmentation result of the UNeXt network adheres two small lesions together, resulting in erroneous segmentation.

Since the COVID-19 dataset contains highly unbalanced categories, the coverage of the lesion area in COVID-19 is relatively small compared with the background, so the ROC curves and the area under it (AUC) were used as the evaluation criteria for the visualization performance of the model. The larger the AUC, the better the model. As shown in Fig. 9, the AUC value of the proposed method is 0.974. Compared with CNN-based networks such as U-Net, Unet++ and CaraNet, the AUC increases by 9.4%, 8.5% and 2.5%, respectively, and compared with 3D Transformer-based networks such as UNETR and Swin-UnetR, the AUC value increases by 8.4% and 8.2%, respectively.

The AUC values increased by 11.3%, 8.9%, 8.1%, 7.9%, 3.5%, 1.9%,

**Fig. 9.** The ROC curves for each model on the COVID-19 dataset.

1.4%, 1.3%, and 0.8%, respectively, when compared to networks based on 2D Transformer types, such as the PVT, SegFormer, TransUnet, Swin-Unet, HiFormer (Moein et al., 2023), Mt-Unet, MedSAM, TransCeption (Reza et al., 2023), and UNeXt.

To analyze the improvement significance of the proposed method

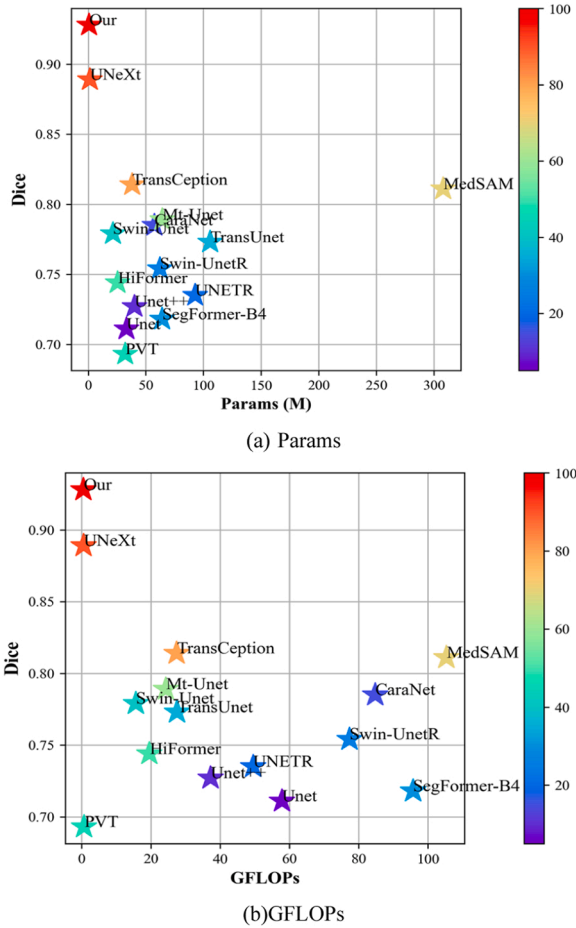


Fig. 10. Visualization of the comparison results for various model parameters and calculations on the COVID-19 dataset.

compared to other methods, the Precision, Recall, Iou, Dice, GFLOPs, and parameter quantities of the test samples were used as inputs, and paired t-tests were used for statistical testing. The P-value in the table is the result of Dice score calculation. It can be observed that the P-value of the proposed method is less than 0.05, indicating a significant improvement * in statistics ($P < 0.05$).

As shown in Fig. 10, on the COVID-19 dataset, the parameter (Params), model calculation amount (GFLOPs), and Dice of each model have been visualized. The MedSAM model not only has a very large number of parameters, but also has a very long calculation time. Due to the

introduction of the Wave Block module in Wave-MLP in this paper to replace the Tok-MLP module, it can be more clearly seen that the Wave Block has fewer parameters than the Tok-MLP module and can significantly improve model performance. The proposed model has fewer parameters and achieves optimal performance by balancing computational complexity and segmentation performance.

3.5.2. Experimental results on the BraTS

According to Table 3, the proposed method achieves optimal segmentation results for three tumor regions on both Dice score and HD distance indicators. The improved model achieved Dice scores of 0.933, 0.925, and 0.918 and HD distances of 1.595, 2.348, and 1.549 on ET, WT, and TC, respectively. Compared to the original model, the Dice scores of the proposed model improved by 4.3%, 3.7%, and 3.9%, while the HD distance decreased by 53.9%, 42.5%, and 37.0%. Compared to the CNN type method and the method based on 2D and 3D Transformer types listed in Table 2, our results obtained are optimal, mainly due to the introduction of the new Wave-MLP module and the addition of the AG module, which can reduce feature loss during the segmentation process, thereby improving model performance. Compared to 3D models, the proposed method also exhibits significant advantages in performance and efficiency, fully demonstrating the effectiveness of the improved model.

The BraTS datasets also have the problem of category imbalance, so this paper also draws ROC curves for each model on the BraTS datasets.

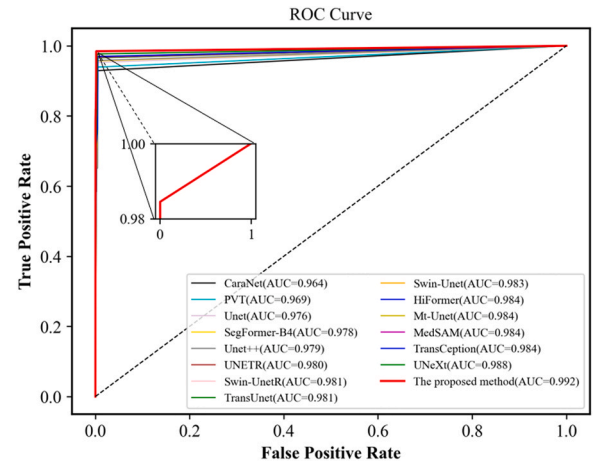


Fig. 11. ROC curves for each model on the BraTS dataset (each ROC curve is plotted using the average value selected from the three types of brain tumor segmentation results).

Table 3

Quantitative comparison of the performance and parameters of the model on the BraTS dataset.

Model Type	Method	Year	Dice Score (%) $\uparrow$			Hausdorff Dist. (mm) $\downarrow$		
			ET	WT	TC	ET	WT	TC
CNN	Unet (Ronneberger et al., 2015)	2015	0.819	0.811	0.777	4.573	6.173	5.363
	Unet+ (Zhou et al., 2018b)	2018	0.842	0.834	0.810	5.436	6.913	4.772
	CaraNet (Ange et al., 2022)	2022	0.791	0.779	0.712	6.381	7.039	6.675
	UNETR (Ali et al., 2022a)	2022	0.863	0.857	0.839	3.650	5.280	3.647
3D Transformer	Swin-NetR (Oktay et al., 2018b)	2022	0.869	0.865	0.849	3.159	5.038	3.340
	SegFormer (Yutong et al., 2021)	2021	0.838	0.823	0.778	4.221	5.637	4.650
	TransUnet (Muhammad et al., 2022)	2021	0.878	0.862	0.851	4.510	5.379	3.362
	Swin-Unet (Cao et al., 2021)	2021	0.880	0.865	0.855	3.578	5.121	3.346
2D Transformer	Mt-Unet (Hongyi et al., 2022)	2022	0.889	0.834	0.872	3.523	4.732	2.873
	PVT (Wang et al., 2021)	2022	0.812	0.801	0.783	4.981	6.843	5.964
	HiFormer (Ali et al., 2022b)	2022	0.881	0.881	0.870	3.366	4.478	2.706
	MedSAM (Junde et al., 2023)	2023	0.889	0.882	0.872	3.383	3.921	2.632
	TransCeption (Moein et al., 2023)	2023	0.889	0.876	0.864	3.687	4.646	2.918
	UNeXt (Rahman and Radu, 2023)	2023	0.890	0.888	0.879	3.466	4.087	2.460
CNN+MLP	Our	–	0.933	0.925	0.918	1.595	2.348	1.549

Each ROC curve in Fig. 11 was plotted using the average value from the three types of brain tumor segmentation results. The performance of each model was represented by the AUC value. The AUC of our method is 1.6%, 1.3% and 2.8% higher than methods based on CNN types, such as U-Net, Unet++ and CaraNet. Compared with 3D segmentation models, it also exhibits excellent performance, which is 1.2% and 1.1% higher than UNETR and Swin-NetR, respectively. At the same time, it is compared with some mainstream Transformer methods at present. Compared with PVT, SegFormer, TransUNet, Swin-Net, HiFormer, Mt-Net, MedSAM, TransCeption, and UNeXt, the AUC has increased by 2.3%, 1.4%, 1.1%, 0.9%, 0.8%, 0.8%, 0.8%, 0.8%, and 0.4%, respectively.

To compare the segmentation results more clearly, the segmentation result is visualized, as shown in Fig. 12. Obviously, compared to U-Net (Ronneberger et al., 2015), MedSAM, and other methods, the segmentation results of the proposed method are closer to Ground truth, with fewer errors and omissions. For segment small tumors well, as some Transformer methods in images with very small tumor regions, some methods cannot Fig. 12 cannot segment details. Our method adds a new loss function, FTL, which attempts to use the γ Coefficients, which are used to learn difficult examples, such as in situations where the ROI is small, and which can more accurately segment brain tumors and produce finer segmentation results.

4. Discussion

In this part, we first carried out ablation experiments to examine how each module affects the performance of the model, followed by model complexity experiments and training strategy research to explore the lightest model. Finally, we conducted model confidence testing and visualized the losses during model training. The optimal performance of the proposed model is verified through a series of experiments.

4.1. Ablation experiments

4.1.1. Ablation Experimental results on the COVID-19

Table 4 shows ablation experiments on the COVID-19 dataset. B represents using the UNeXt network as the baseline network, and B+Loss represents that B adds a new loss function 2.2%, 3.4%, 2.4%, and 1.9% for Dice, Iou, Recall, and Precision, respectively. B+Loss+AG indicates that adding an AG module at the skip connection has achieved greater performance improvement. Compared to the baseline model B, it

has achieved an average growth of 3.4%, 5.7%, 3.3%, and 3.6% in Dice, Iou, Recall, and Precision, respectively, and the parameter of the model has not increased significantly. Because the AG module can fuse feature loss during the down-sampling process, thereby improving model performance. B+Loss+AG+ the Wave block indicates that Tok-MLP is replaced with the Wave block in Wave-MLP, which not only improves the performance but also reduces the parameter by 48.2%, achieving the goal of a lightweight model.

At the same time, T-tests were conducted on the improved model to verify the effectiveness of the improved method. The P-value in the table is the result of the Dice score calculation. Meanwhile, the P-values of other indicators are all less than 0.05, indicating that the improvement in this article contributed significantly to the original model.

4.1.2. Ablation Experimental Results on the BraTS

As shown in Table 5, an ablation experiment was conducted on a BraTS dataset using a Dice score and HD distance for quantitative evaluation. The baseline network is UNeXt, and three improvements have been made. The added modules provide a richer feature representation for the proposed method. Specifically, the Dice scores of ET, WT, and TC increased by 2.1%, 1.8%, and 2.0%, respectively, compared to the three categories of original network brain tumors, while the HD distance decreased by 27.5%, 22.1%, and 25.2%, respectively, indicating that adding a new loss function FTL γ coefficients are used to learn difficult examples, such as when the ROI (region of interest) is small, which can more accurately segment brain tumors and produce more refined segmentation results. An AG module was added at the skip connection. Compared to the baseline model baseline, the Dice scores of ET, WT, and TC were increased by 3.9%, 3.3%, and 3.7%, respectively, and the HD distance was reduced by 49.6%, 32.8%, and 31.1%, respectively. It reduced information loss during feature fusion, resulting in a significant performance improvement, and the AG module did not add any additional burden of computational complexity. B+Loss+AG+Wave-MLP indicates that in addition to performing the above operations on the network, Tok-MLP in the original network was replaced with the Wave Block module in Wave-MLP which achieving a balance between the accuracy of the model and computational complexity.

4.2. Model complexity experiment

The improved model convolution stage uses the UNeXt structure, and

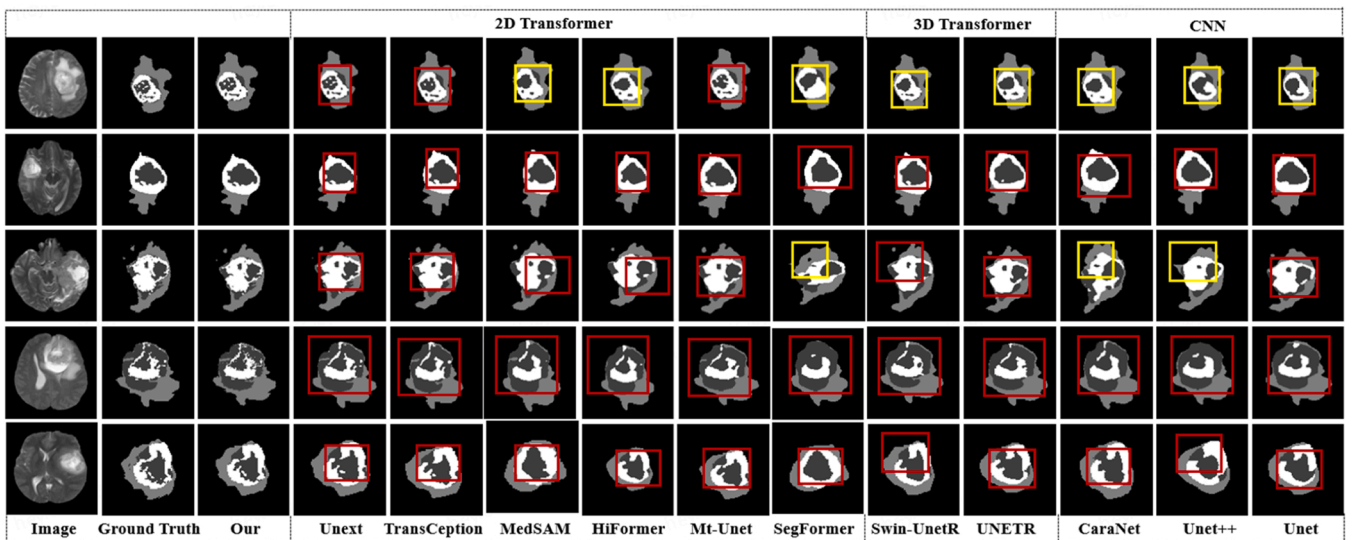


Fig. 12. Comparison diagram of model segmentation results on the BraTS dataset (The red box indicates a detail segmentation error, and the yellow box indicates a missing segmentation).

Table 4

Ablation experiments with different modules on the COVID-19 dataset.

Method	Dice	Iou	Recall	Precision	P	Parmars (M)
B	0.889	0.803	0.881	0.897	0.003	1.47
B+Loss	0.911	0.837	0.905	0.933	0.004	1.47
B+Loss+AG	0.923	0.860	0.914	0.933	0.015	1.59
B+Loss+AG+Wave block	0.928	0.867	0.916	0.940	-	0.76

Table 5

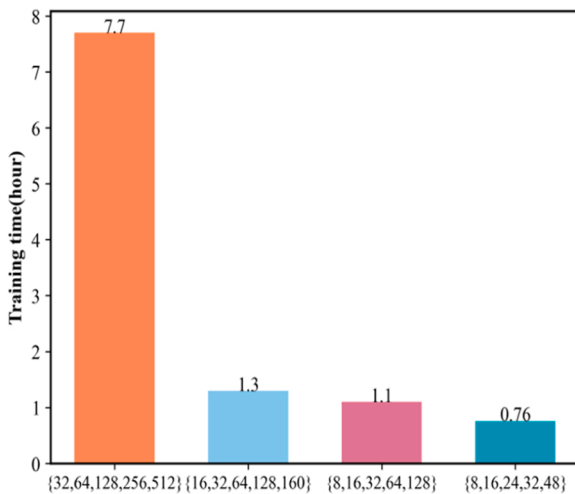
Ablation experiments with different modules on the BraTS dataset.

Method	Dice Score (%) ↑			HD (%) ↓			Parmars (M)
	ET	WT	TC	ET	WT	TC	
B	0.890	0.888	0.879	3.466	4.087	2.460	1.47
B+Loss	0.911	0.906	0.899	2.510	3.181	1.838	1.47
B+Loss+AG	0.929	0.921	0.916	1.744	2.746	1.694	1.59
B+Loss+AG+Wave-MLP	0.933	0.925	0.918	1.595	2.348	1.549	0.76

Table 6

Improved model using different channel training times on the BraTS datasets.

Method	Dice Score (%) ↑			HD (%) ↓			Parmars (M)
	ET	WT	TC	ET	WT	TC	
{32,64,128,256,512}	0.922	0.913	0.907	1.875	2.340	1.545	7.7
{16,32,64,128,160}	0.925	0.918	0.907	1.908	2.219	1.499	1.3
{8,16,32,64,128}	0.925	0.922	0.914	1.650	2.122	1.635	1.1
{8,16,24,32,48}	0.933	0.925	0.918	1.595	2.348	1.549	0.76

**Fig. 13.** Visualization of training time using different channels on BraTS datasets using improved models.

the pure convolution quantity of each stage is one. In the original paper, the number of channels in each stage was {32,64,128,256,512}. To reduce the parameters of the model, a discussion experiment on the number of channels was conducted to evaluate their impact on performance and the number of trainable parameters. As shown in Table 6, when the number of channels is set to {8,16,24,32,48}, the model parameters are minimal and the performance is relatively optimal.

As shown in the visualization in Fig. 13 below, if it is reduced to

{8,16,24,32,48}, the number of trainable parameters is one tenth of the number of parameters in the original model, while the overall performance is still equivalent. According to the comparison of experimental results, the {8,16,24,32,48} configuration was used in all experiments in this article.

4.3. Training Strategy Experiment

To reduce training time, various training strategies were investigated, and the best training method was chosen. (1) Individual training occurs when training candidate networks. (2) Joint training simultaneously trains all candidate networks. Table 7 below shows the performance and training time comparisons of the two training strategies for segmenting BraTS datasets. Joint training not only greatly reduces training time, but also achieves higher performance. Compared to Individual training, Joint training increased Dice by 1.3%, 1.4%, and 1.2%, decreased HD by 24.5%, 8.7%, and 6.4%, and shortened training time by 48.9%.

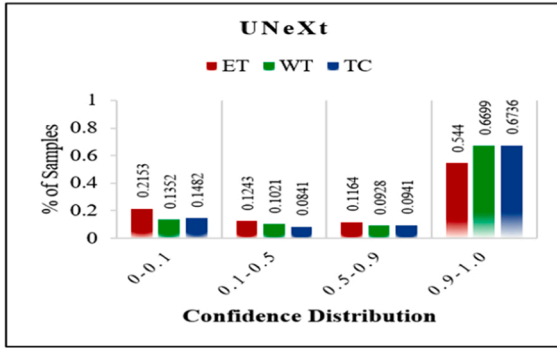
4.4. Confidence experiment

This paper compares the prediction confidence distribution of UNExT with the proposed method. As a separate radius measurement, the confidence interval is commonly referred to as the error range. The larger the estimated sample size, the more accurate the estimation and the narrower the confidence interval. Because the prediction probability is mostly distributed within the ranges of [0,0.1] and [0.9,1.0], the intermediate interval ([0.1,0.9]) is split into two intervals, [0.1,0.5] and [0.5,0.9], to simplify processing. As shown in Fig. 14, compared to

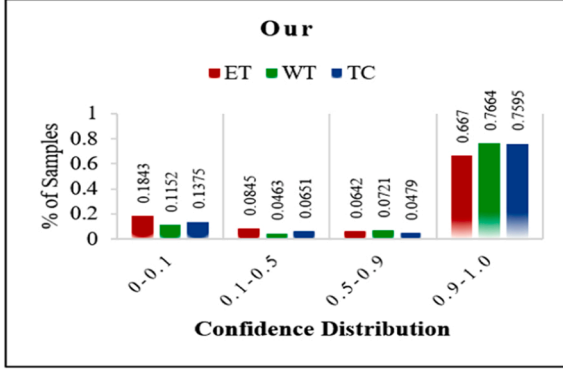
Table 7

Ablation study on the impact of different training strategies on candidate networks.

Method	Dice Score (%) ↑			HD (%) ↓			Training time (Hour)
	ET	WT	TC	ET	WT	TC	
Individual training	0.920	0.911	0.906	2.113	2.572	1.655	10.47
Joint training	0.933	0.925	0.918	1.595	2.348	1.549	5.34



(a)UNeXt



(b)Our Model

Fig. 14. Comparison of confidence distributions for model predictions based on a five-fold cross validation evaluation of training sets from the BraTS dataset.

UNeXt, the prediction probability of the proposed method falls more frequently within [0.9,1.0], and the segmentation results for the three categories are improved by 12.3%, 9.6%, and 8.5%, respectively, with higher prediction confidence. In addition, the intermediate interval ([0.1,0.9]) further proves that the proposed model in this paper can discard uncertain predictions and make more accurate predictions compared to the original UNeXt.

4.5. Training process validation

Fig. 15 (a) shows the loss curve during training and validation. At the beginning of training, when Epoch is 0–50, the loss function value of the

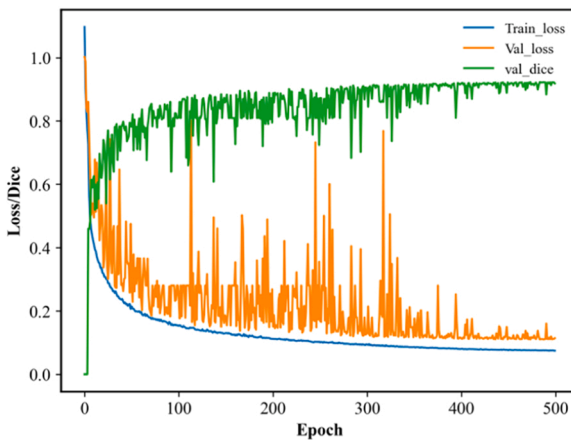


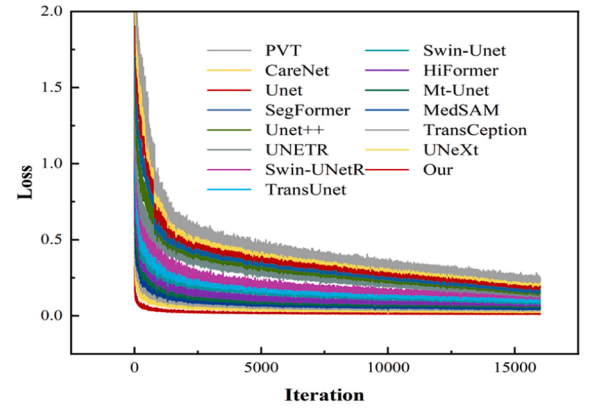
Fig. 15. Comparison of the loss curve during training and Dice curve during verification on the COVID-19 dataset.

training set converges relatively quickly, but the loss fluctuation of the validation set is relatively large, indicating that the selected learning rate is relatively suitable. After learning to a certain stage, the loss curve tends to stabilize when the epoch is between 400 and 500, and the loss change is not as obvious as at the end, indicating that it tends to converge. **Fig. 15** (b) shows the Dice curve during the validation process. During the epoch period of 400 and 500, the Dice variation tends to stabilize, approaching 0.9 and achieving the best performance.

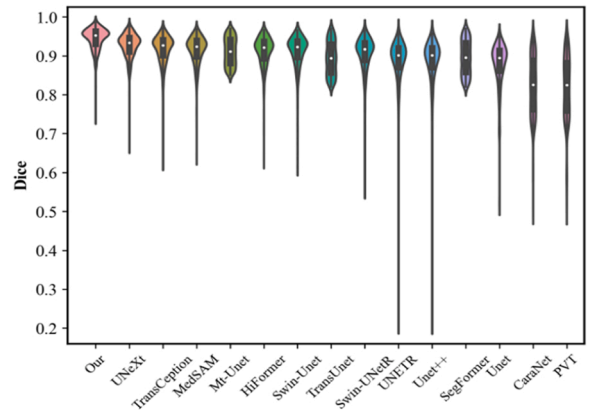
Fig. 16 (a) shows the loss curve during the training process of various models on the BraTS dataset. The proposed method converges to the minimum first and decreases the fastest. At the same time, the Dice values of the test samples were redrawn using a violin chart. As shown in **Fig. 16** (b), the lowest Dice value of the proposed model is between 0.7 and 0.8, and the Dice values of most samples are stable between 0.9 and 1.0. The proposed model has excellent performance, while other models such as UNETR and Swin-UnetR have examples with Dice values of 0.2, indicating that the segmentation model is unstable.

5. Conclusion

This paper proposed a lightweight network model for medical image segmentation. Extensive experimental results on two datasets (COVID-19 CT and BraTS 2018 MRI) show that the proposed method is more competitive than previous advanced methods. And it can make up for the two shortcomings proposed in the introduction. The improved model Dice score, Precision score, Recall score, and Iou score in the COVID-19 CT dataset are 0.928, 0.867, 0.916, and 0.940, respectively, and the parameter amount is the lowest, with GLOPs and Params being 0.52 and 0.76, respectively. On the BraTS 2018 MRI dataset, the Dice



(a)Loss



(b)Dice

Fig. 16. Loss curve during training and Dice curve during testing of each model on the BraTS dataset.

scores for ET, WT, and TC were 0.933, 0.925, and 0.918, respectively, and the HD scores for ET, WT, and TC are 1.595, 2.348, and 1.549, respectively. In future work, we shall attempt to explore how to apply two-dimensional (Mamalakidis et al., 2021) segmentation models to guide practical research. Attempts are also being made to investigate the visualization of three-dimensional reconstruction and two-dimensional segmentation results to better assist doctors in screening, diagnosing (Alhasan and Hasaneena, 2021), and tracking treatment.

CRedit authorship contribution statement

Yi He: Conceptualization, Writing – original draft, Methodology, Visualization, Formal analysis, Software, Validation, Writing – review & editing. **Zhijun Gao:** Visualization, Funding acquisition, Data curation, Writing – review&editing, Project administration, Investigation. **Yi Li:** Writing – review & editing, Formal analysis, Conceptualisation. **Zhiming Wang:** Visualization, Investigation.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Publicly available datasets were analyzed in this study. COVID-19: <https://www.kaggle.com/datasets/andrewmvd/covid19-ct-scans>.

BraTS: MICCAI BRATS - The Multimodal Brain Tumor Segmentation Challenge.

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Publicly available datasets were analyzed in this study

These data can be found here: COVID-19: <https://www.kaggle.com/datasets/andrewmvd/covid19-ct-scans>.

BraTS: MICCAI BRATS - The Multimodal Brain Tumor Segmentation Challenge.

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